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## Introduction to Cryptography

A1. The report within Scenario 1 details that BrightFuture will implement a multitude of changes, including clerk payments, which is used to process things such as permit and tax payments by using government workstations. The way these transactions are verified is by authenticating the identity of a clerk or citizen, and the integrity aspect is covered by ensuring the dates and payment amounts are accurate and not tampered with. Neither party may deny the other.

All employees can use their own personal devices to send emails. The way that this is being authenticated is by confirming the sender's address, the integrity is covered by ensuring all the content being sent is secure and unchanged during transmission, and the nonrepudiation is done by leaving a digital signature made from the sender, which also counts as accountability for anything sent from the sender's address.

As for the law enforcement stored body camera footage, this will be kept within the data center of the agency. It'll be authenticated by devices used to capture the footage in the first place. Integrity is covered by ensuring the footage was not edited and is in its original state when stored. Nonrepudiation is preserved through the chain of custody, so the source of the footage cannot be denied if it is used as evidence.

A2. The financial and payment data of citizens must be kept encrypted in a format that keeps the data unreadable from anyone who doesn't have access to the key, not only at rest but when it's in transmission as well. The only way for it to be accessed is by staff or systems with authorization. The main weakness that I identified is with the cryptographic solution being required because plaintext payment data can easily be read and intercepted, which could expose sensitive customer information, such as taxpayer and card information.

As for the camera footage encryption supports it through integrity, because authenticated encryption lets the city detect any change, so any footage that is accessed is provably unaltered. The main weakness of this is that the footage is unprotected and can be altered, which would affect the value of the evidence.

A3. The three examples that I noticed were vulnerabilities based on scenario 1. The first is regarding the email system, as it relies entirely on the RSA, which requires to be passed through PKI. The problem here is that strength in itself is entirely reliant on the length of the key if it's short, it's more susceptible to being weak and vulnerable (zyBooks, 2025). Because of this, it is more susceptible to being vulnerable to brute force attacks and chosen-ciphertext attacks. Two protocols that would be a great solution for this would be S/MIME, as it encrypts and signs every message that passes through the network with RSA and AES, which makes any mail that's intercepted completely unreadable. If an email was tampered with, it would be detectable. The other protocol that would help is adding a TLS layer to secure the overall delivery channel with ephemeral keys and strong ciphers which prevent interceptions from happening during transmission.

As for the payment processing a vulnerability that I identified was that it has a fallback, which leads to weak cipher suites that are completely exposed and are susceptible to brute force and known plaintext attacks. The two protocols that would be a good solution to this are TLS encryption and IPsec. The TLS encryption adds strong cipher suites and it also makes brute force virtually suboptimal. IPsec would encrypt and add an authentication layer to the traffic moving through the network, and it would protect data even if things such as the application session may be misconfigured.

The camera footage that is being stored is also at risk of attacks. The vulnerability that I noticed is that it has large volumes being placed within a reused key that has been used for a long time. It's currently exposed to two forms of attacks, ciphertext-only and brute force attacks. The solution to fix this is to use two protocols that tackle the vulnerability at its heart. The first protocol would be IPsec, as this protects the stored footage when it's in transmission by using symmetric encryption. The second protocol would be the SSH/SFTP which completely encrypts bulk transfers as well as any access to the storage too, it also reduces the exposure of the large cipher text from attackers keeping it hidden and secure even with so much data under one key.

A4. The first example of balancing the relationship between performance and security within scenario 1 is the IoT sensors. The IoT sensors such as the environmental, power grid, and traffic devices, they are all performance based, focused on low power and saving costs, so implementing a heavy encryption system for every transmission would not only affect performance but slow down the real time data the city depends on. The balance for this is using lightweight symmetric encryption and session keys instead of relying on high cost public keys.

The second example is the citizen portal servers. As these focus on scalability, they have large numbers of users, so they tend to use full PKI and TLS protocol, which balance the security against scalability, because the city relies on both efficiency and security, which is why the usage of hardware-accelerated encryption is optimal, because cryptography won't become a bottleneck as the city grows.

A5. The first cryptanalytic methods that would be able to break the cryptographic system is brute force attacks, as this tries every possible combination of keys until it's able to decrypt the data. But the chances of this being effective entirely depend on the weakness of a key, the key length. The longer it is the harder it is to break as it takes more time and is the reason why most modern systems require long keys.

The second cryptanalytic method that would be able to break the cryptographic system is frequency analysis as this is able to conduct an analysis of patterns within the cipher text and see how often certain blocks or characters appear and map those patterns to what the plaintext most likely would be. It's mostly only effective against ciphers that contain weak, predictable patterns but it fails against the modern strong algorithms that have no noticeable structure patterns.

## **References**

zyBooks. (2025). *WGU D830: Introduction to cryptography* (Version 1).

<https://learn.zybooks.com/zybook/WGUD830Cryptographyv1>